

F4 Laws of logarithms

Name: _____

Class: _____

Date: _____

Time: **261 minutes**

Marks: **220 marks**

Comments:

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Q1.

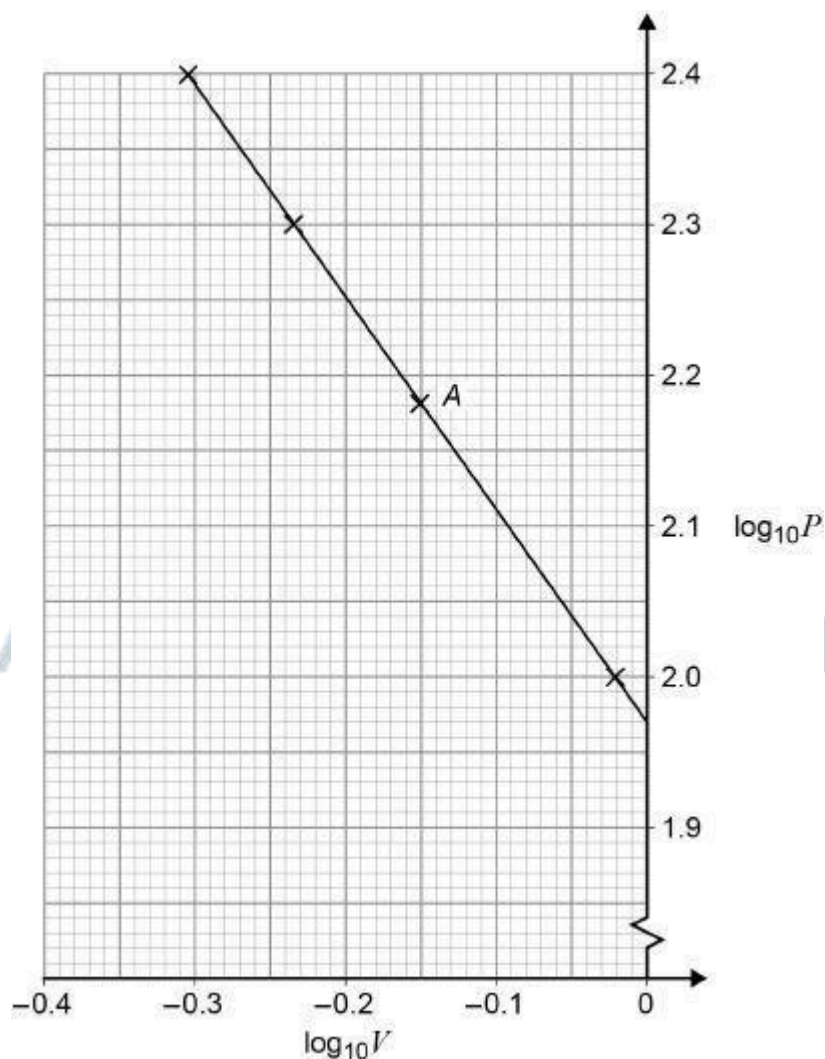
Maxine measures the pressure, P kilopascals, and the volume, V litres, in a fixed quantity of gas.

Maxine believes that the pressure and volume are connected by the equation

$$P = cV^d$$

where c and d are constants.

Using four experimental results, Maxine plots $\log_{10} P$ against $\log_{10} V$, as shown in the graph below.



(a) Find the value of P and the value of V for the data point labelled A on the graph.

(2)

(b) Calculate the value of each of the constants c and d .

(4)

- (c) Estimate the pressure of the gas when the volume is 2 litres.

(2)

(Total 8 marks)

Q2.

Express as a single logarithm

$$2 \log_a 6 - \log_a 3$$

(Total 2 marks)

Q3.

- (a) Given then $\log_a y = 2 \log_a 7 + \log_a 4 + \frac{1}{2}$, find y in terms of a .

(4)

- (b) When asked to solve the equation

$$2 \log_a x = \log_a 9 - \log_a 4$$

a student gives the following solution:

$$2 \log_a x = \log_a 9 - \log_a 4$$

$$\Rightarrow 2 \log_a x = \log_a \frac{9}{4}$$

$$\Rightarrow \log_a x^2 = \log_a \frac{9}{4}$$

$$\Rightarrow x^2 = \frac{9}{4}$$

$$\therefore x = \frac{3}{2} \text{ or } -\frac{3}{2}$$

Explain what is wrong with the student's solution.

(1)

(Total 5 marks)

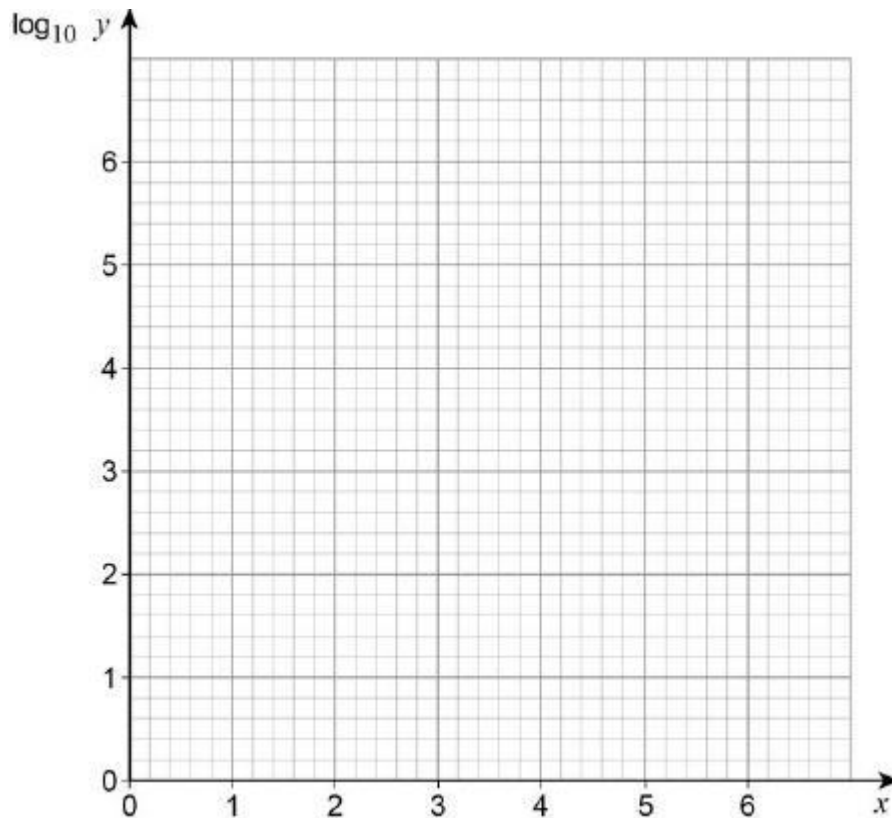
Q4.

A student conducts an experiment and records the following data for two variables, x and y .

x	1	2	3	4	5	6
y	14	45	130	1100	1300	3400
$\log_{10} y$						

The student is told that the relationship between x and y can be modelled by an equation of the form $y = kb^x$

- (a) Plot values of $\log_{10} y$ against x on the grid below.



- (b) State, with a reason, which value of y is likely to have been recorded incorrectly.

- (c) By drawing an appropriate straight line, find the values of k and b .

(2)
(1)
(4)
(Total 7 marks)

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Q5.

Find the value of $\log_a(a^3) + \log_a\left(\frac{1}{a}\right)$

(Total 2 marks)

Q6.

A circular ornamental garden pond, of radius 2 metres, has weed starting to grow and cover its surface.

As the weed grows, it covers an area of A square metres. A simple model assumes that the weed grows so that the rate of increase of its area is proportional to A .

- (a) Show that the area covered by the weed can be modelled by

$$A = Be^{kt}$$

where B and k are constants and t is time in days since the weed was first noticed.

(4)

- (b) When it was first noticed, the weed covered an area of 0.25 m^2 . Twenty days later the weed covered an area of 0.5 m^2

(i) State the value of B .

(1)

(ii) Show that the the model for the area covered by the weed can be written as

$$A = 2^{\frac{t}{20} - 2}$$

(4)

(iii) How many days does it take for the weed to cover half of the surface of the pond?

(2)

(c) State one limitation of the model.

(1)

(d) Suggest one refinement that could be made to improve the model.

(1)

(Total 13 marks)

Q7.

Given that

$$\log_a N - \log_a x = \frac{3}{2}$$

express x in terms of a and N , giving your answer in a form not involving logarithms.

(Total 3 marks)

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Q8.

(a) Given that $\log_a b = c$, express b in terms of a and c .

(1)

(b) By forming a quadratic equation, show that there is only one value of x which satisfies the equation $2 \log_2(x + 7) - \log_2(x + 5) = 3$.

(6)

(Total 7 marks)

Q9.

(a) Sketch the graph of $y = \frac{1}{2^x}$, indicating the value of the intercept on the y -axis.

(2)

- (b) Use logarithms to solve the equation $\frac{1}{2^x} = \frac{5}{4}$, giving your answer to three significant figures. (3)

- (c) Given that

$$\log_a(b^2) + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a}\right)$$

express y in terms of a and b .

Give your answer in a form not involving logarithms.

(5)

(Total 10 marks)

Q10.

- (a) Given that $2 \log_k x - \log_k 5 = 1$, express k in terms of x . Give your answer in a form not involving logarithms.

(4)

- (b) Given that $\log_a y = \frac{3}{2}$ and that $\log_a a = b + 2$, show that $y = 2^p$, where p is an expression in terms of b .

(3)

(Total 7 marks)

Q11.

The first term of a geometric series is 12 and the common ratio of the series is $\frac{3}{8}$.

- (a) Find the sum to infinity of the series. (2)

- (b) Show that the sixth term of the series can be written in the form $\frac{3^6}{2^{13}}$. (3)

- (c) The n th term of the series is u_n .

- (i) Write down an expression for u_n in terms of n . (1)

- (ii) Hence show that

$$\log_a u_n = n \log_a 3 - (3n - 5) \log_a 2$$

(4)

(Total 10 marks)

Q12.

- (a) Given that $3 \ln x = 4$, find the exact value of x .

(1)

- (b) By forming a quadratic equation in $\ln x$, solve $3 \ln x + \frac{20}{\ln x} = 19$, giving your answers for x in an exact form.

(5)

(Total 6 marks)

Q13.

The variables x and Y , where $Y = \log_{10} y$, are related by the equation

$$Y = mx + c$$

where m and c are constants.

- (a) Given that $y = ab^x$, express a in terms of c , and b in terms of m .

(3)

- (b) It is given that $y = 12$ when $x = 1$ and that $y = 27$ when $x = 5$.

On the diagram below, draw a linear graph relating x and Y .

(3)

- (c) Use your graph to estimate, to two significant figures:

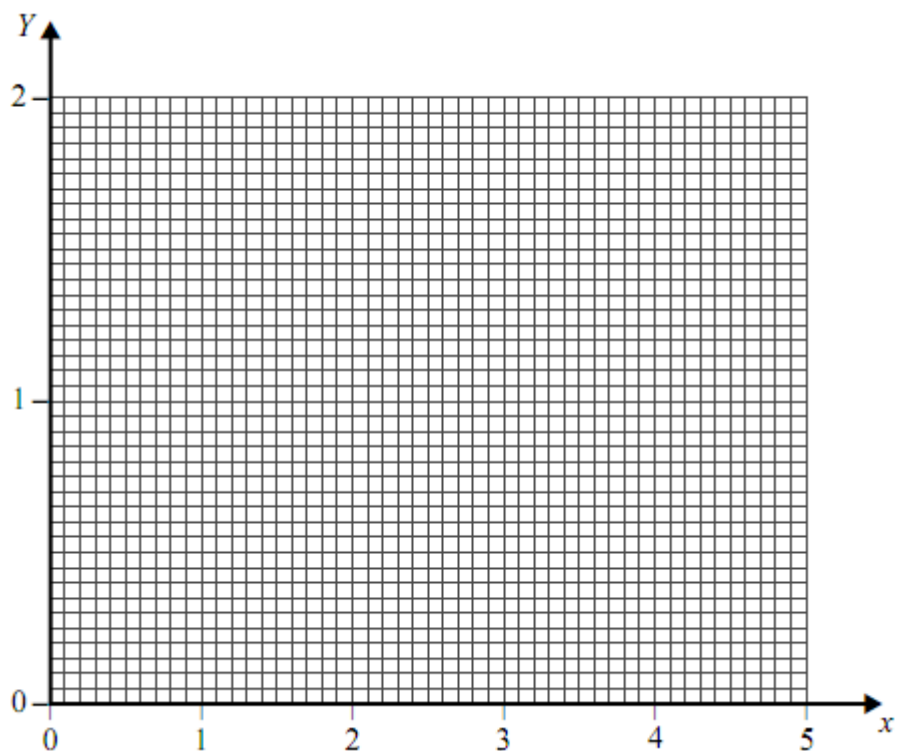
- (i) the value of y when $x = 3$;

(2)

- (ii) the value of a .

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(2)
(Total 10 marks)

Q14.

(a) Find the value of x in each of the following:

(i) $\log_9 x = 0$;

(1)

(ii) $\log_9 x = \frac{1}{2}$.

(1)

(b) Given that

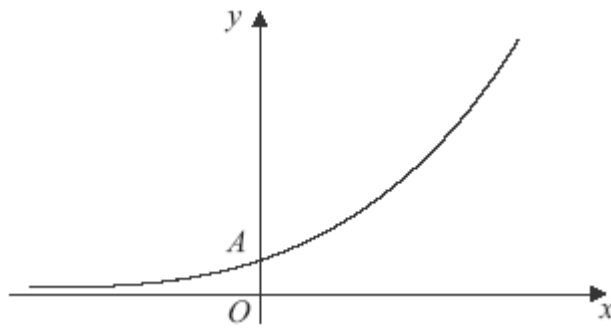
$$2 \log_a n = \log_a 18 + \log_a (n - 4)$$

find the possible values of n .

(5)
(Total 7 marks)

Q15.

The diagram shows a sketch of the curve $y = 2^{4x}$.



The curve intersects the y -axis at the point A .

- (a) Find the value of the y -coordinate of A . (1)

- (b) Use the trapezium rule with six ordinates (five strips) to find an approximate value for $\int_0^1 2^{4x} \, dx$, giving your answer to two decimal places. (4)

- (c) Describe the geometrical transformation that maps the graph of $y = 2^{4x}$ onto the graph of $y = 2^{4x-3}$. (2)

- (d) The curve $y = 2^{4x}$ is translated by the vector $\begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$ to give the curve $y = g(x)$.
The curve $y = g(x)$ crosses the x -axis at the point Q . Find the x -coordinate of Q . (4)

- (e) (i) Given that

$$\log_a k = 3 \log_a 2 + \log_a 5 - \log_a 4$$

show that $k = 10$. (3)

- (ii) The line $y = \frac{5}{4}$ crosses the curve $y = 2^{4x-3}$ at the point P . Show that the

x -coordinate of P is $\frac{1}{4 \log_{10} 2}$. (3)

(Total 17 marks)

Q16.

- (a) Write each of the following in the form $\log_a k$, where k is an integer:
- (i) $\log_a 4 + \log_a 10$; (1)
- (ii) $\log_a 16 - \log_a 2$; (1)
- (iii) $3 \log_a 5$. (1)
- (b) Use logarithms to solve the equation $(1.5)^{3x} = 7.5$, giving your value of x to three decimal places. (3)
- (c) Given that $\log_2 p = m$ and $\log_8 q = n$, express pq in the form 2^y , where y is an expression in m and n . (3)
- (Total 9 marks)**

Q17.

- (a) (i) Find the value of p for which $\sqrt{125} = 5^p$. (2)
- (ii) Hence solve the equation $5^{2x} = \sqrt{125}$. (1)
- (b) Use logarithms to solve the equation $3^{2x-1} = 0.05$, giving your value of x to four decimal places. (3)
- (c) It is given that
- $$\log_a x = 2(\log_a 3 + \log_a 2) - 1$$
- Express x in terms of a , giving your answer in a form not involving logarithms. (4)
- (Total 10 marks)**

Q18.

- (a) Given that
- $$\log_a x = \log_a 16 - \log_a 2$$
- write down the value of x . (1)
- (b) Given that

$$\log_a y = 2 \log_a 3 + \log_a 4 + 1$$

express y in terms of A , giving your answer in a form **not** involving logarithms.

(3)

(Total 4 marks)

Q19.

- (a) (i) Given that $y = 2x^2 - 8x + 3$, find $\frac{dy}{dx}$.

(1)

- (ii) Hence, or otherwise, find

$$\int_4^6 \frac{x-2}{2x^2-8x+3} dx$$

giving your answer in the form $k \ln 3$, where k is a rational number.

(4)

- (b) Use the substitution $u = 3x - 1$ to find , giving your answer in terms of x .

(4)

(Total 9 marks)

Q20.

- (a) Given that $\frac{3}{9-x^2}$ can be expressed in the form $k\left(\frac{1}{3+x} + \frac{1}{3-x}\right)$, find the value of the rational number k .

(2)

- (b) Show that $\int_1^2 \frac{3}{9-x^2} dx = \frac{1}{2} \ln\left(\frac{a}{b}\right)$, where a and b are integers.

(3)

(Total 5 marks)

Q21.

- (a) Write down the value of:

(i) $\log_a 1$;

(1)

(ii) $\log_a a$.

(1)

- (b) Given that

$$\log_a x = \log_a 5 + \log_a 6 - \log_a 1.5$$

find the value of x .

(3)

(Total 5 marks)

Q22.

- (a) Solve the equation $3 \log_a x = \log_a 8$.

(2)

- (b) Show that

$$3 \log_a 6 - \log_a 8 = \log_a 27$$

(3)

- (c) (i) The point $P(3, p)$ lies on the curve $y = 3 \log_{10} x - \log_{10} 8$.

Show that
$$p = \log_{10} \left(\frac{27}{8} \right).$$

(2)

- (ii) The point $Q(6, q)$ also lies on the curve $y = 3 \log_{10} x - \log_{10} 8$.

Show that the gradient of the line PQ is $\log_{10} 2$.

(4)

(Total 11 marks)

Q23.

The variables x and y are related by an equation of the form

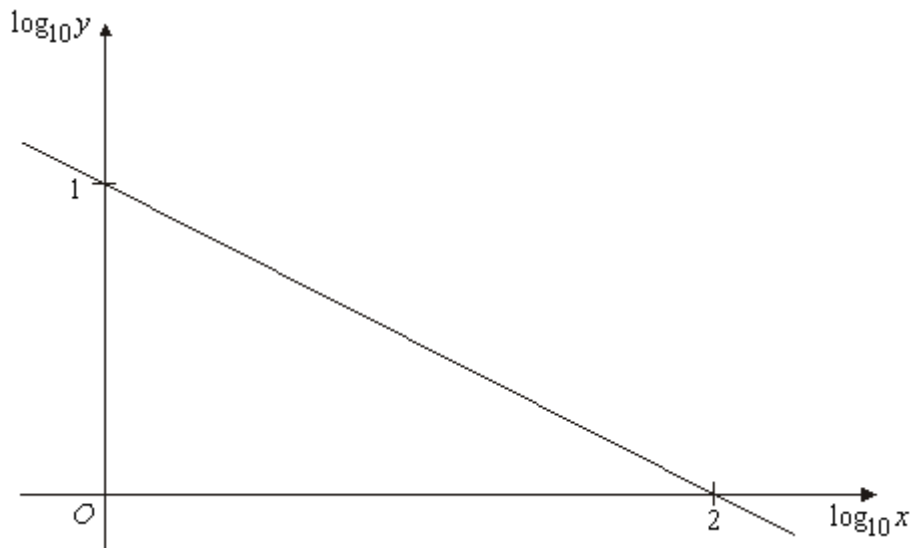
$$y = ax^b$$

where a and b are constants.

- (a) Using logarithms to base 10, reduce the relation $y = ax^b$ to a linear law connecting $\log_{10} x$ and $\log_{10} y$.

(2)

- (b) The diagram shows the linear graph that results from plotting $\log_{10} y$ against $\log_{10} x$.



Find the values of a and b .

(4)

(Total 6 marks)

Q24.

- (a) It is given that n satisfies the equation

$$\log_a n = \log_a 3 + \log_a (2n - 1)$$

Find the value of n .

(3)

- (b) Given that $\log_a x = 3$ and $\log_a y - 3 \log_a 2 = 4$:

- (i) express x in terms of a ;

(1)

- (ii) express xy in terms of a .

(4)

(Total 8 marks)

Q25.

It is given that n satisfies the equation

$$2 \log_a n - \log_a (5n - 24) = \log_a 4$$

- (a) Show that $n^2 - 20n + 96 = 0$.

(3)

- (b) Hence find the possible values of n .

(2)

(Total 5 marks)

Q26.

- (a) Given that

$$\log_a x = 2 \log_a 6 - \log_a 3$$

show that $x = 12$.

(3)

- (b) Given that

$$\log_a y + \log_a 5 = 7$$

express y in terms of a , giving your answer in a form not involving logarithms.

(3)

(Total 6 marks)

Q27.

- (a) Given that

$$\log_a x = 3 \log_a 6 - \log_a 8$$

where a is a positive constant, show that $x = 27$.

(3)

- (b) Write down the value of:

(i) $\log_4 1$;

(1)

(ii) $\log_4 4$;

(1)

(iii) $\log_4 2$;

(1)

(iv) $\log_4 8$.

(1)

(Total 7 marks)

Q28.

The sum to infinity of a geometric series is four times the first term of the series.

- (a) Show that the common ratio, r , of the geometric series is $\frac{3}{4}$.

(3)

- (b) The first term of the geometric series is 48. Find the sum of the first 10 terms of the series, giving your answer to four decimal places.

(2)

(c) The n th term of the geometric series is u_n and the $(2n)$ th term of the series is u_{2n} .

(i) Write u_n and u_{2n} in terms of n .

(2)

(ii) Hence show that $\log_{10}(u_n) - \log_{10}(u_{2n}) = n \log_{10} \left(\frac{4}{3} \right)$.

(3)

(iii) Hence show that the value of $\log_{10} \left(\frac{24_{100}}{24_{200}} \right)$ is 12.5 correct to three significant figures.

(2)

(Total 12 marks)

Q29.

(a) Using the binomial expansion, or otherwise, express $(1 + x)^4$ in ascending powers of x .

(3)

(b) (i) Hence show that $(1 + \sqrt{5})^4 = 56 + 24\sqrt{5}$.

(3)

(ii) Hence show that $\log_2(1 + \sqrt{5})^4 = k + \log_2(7 + 3\sqrt{5})$, where k is an integer.

(3)

(Total 9 marks)